Lecture 03.07: Practice Examples (Loops)

📌 Executive Summary

Hands-on while loop patterns: Counter sequences, cumulative accumulators, sentinel input loops, and digit extraction.

🔬 Theoretical Foundations & Principles

1. Canonical Loop Design Patterns

- **Counter-Controlled Loop:** Iterates a fixed number of times based on an integer counter.
- **Accumulator Pattern:** Progressively combines sequence elements into a running aggregate.
- **Sentinel Loop:** Continues indefinitely until a specific marker value is encountered.
- **Successive Division (Digit Deconstruction):** Uses $N \pmod{10}$ and $N//10$ to inspect digits in integer arithmetic.

💻 Production Implementation & Code

```
# Digit extraction pattern via while loop
def extract_digits_reversed(n: int) -> list[int]:
    digits = []
    temp = abs(n)
    if temp == 0:
        return [0]
    while temp > 0:
        digit = temp % 10
        digits.append(digit)
        temp //= 10
    return digits

print("Digits of 48392:", extract_digits_reversed(48392))
```

⚙️ Algorithmic Complexity & Systems Analysis

- **Time Complexity:** Detailed in respective algorithmic operations.

- **Auxiliary Space:** Memory bounds and CPython frame management.